Handout on Geometric and P-Series Convergence Tests

Part I: Practice Questions

1. Determine if the series converges or diverges. If it converges, find its sum:

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{4^n}$$

2. Determine if the series converges or diverges. If it converges, find its sum:

$$\sum_{n=1}^{\infty} \left(1 - \frac{7}{4^n}\right)$$

3. Determine if the series converges or diverges. If it converges, find its sum:

$$\sum_{n=0}^{\infty} (-1)^n \frac{5}{4^n}$$

4. Determine if the series converges or diverges. If it converges, find its sum:

$$\sum_{n=0}^{\infty} \left(\frac{5}{2^n} + \frac{1}{3^n}\right)$$

5. Determine if the series converges or diverges. If it converges, find its sum:

$$1 + (-3) + (-3)^2 + (-3)^3 + \dots$$

6. Determine if the series converges or diverges. If it converges, find its sum:

$$\left(-\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)^3 + \left(-\frac{2}{3}\right)^4 + \dots$$

7. Test the nature of the series:

$$\sum_{n=1}^{\infty} \frac{n}{n+10}$$

8. Test the nature of the series:

$$\sum_{n=1}^{\infty} \frac{n}{n^2+3}$$

9. Test the nature of the series:

$$\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$$

10. Test the nature of the series:

$$\sum_{n=0}^{\infty} \cos(n\pi)$$

11. Find a formula for the n-th partial sum S_n and use it to determine if the series converges or diverges:

$$\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

12. Find a formula for the n-th partial sum S_n and use it to determine if the series converges or diverges:

$$\sum_{n=1}^{\infty} \left[\cos^{-1}\left(\frac{1}{n+1} \right) - \cos^{-1}\left(\frac{1}{n+2} \right) \right]$$

13. Find a formula for the n-th partial sum S_n and use it to determine if the series converges or diverges:

$$\sum_{n=1}^{\infty} \left(\frac{3}{n^2} - \frac{3}{(n+1)^2} \right)$$

14. Find the sum of the series:

$$\sum_{n=1}^{\infty} \frac{4}{(4n-3)(4n+1)}$$

15. Find the sum of the series:

$$\sum_{n=1}^{\infty} \frac{40n}{(2n-1)^2(2n+1)^2}$$

16. Find the sum of the series:

$$\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right)$$

17. Can you make an infinite series of non-zero terms that converges to any number you want? Explain.

18. If $\sum a_n$ converges and $a_n > 0$ for all n, can anything be said about $\sum \frac{1}{a_n}$? Give reasons for your answer.

19. Are there any values of $x \neq 0$ for which $\sum_{n=1}^{\infty} \frac{1}{nx}$ converges? Give reasons.

20. Is it true that if $\sum a_n$ is a divergent series of positive numbers, then there is also a divergent series $\sum b_n$ of positive numbers with $b_n < a_n$ for every n?

21. Use a comparison test to determine the nature of the series:

$$\sum_{n=1}^{\infty} \frac{1}{n^2 + 30}$$

22. Use a comparison test to determine the nature of the series:

$$\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^{3/2}}$$

23. Use a comparison test to determine the nature of the series:

$$\sum_{n=1}^{\infty} \frac{1}{n \cdot 3^n}$$

24. Use a comparison test to determine the nature of the series:

$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$

25. Use a comparison test to determine the nature of the series:

$$\sum_{n=2}^{\infty} \frac{n(n+1)}{(n^2+1)(n-1)}$$

26. Use a comparison test to determine the nature of the series:

$$\sum_{n=1}^{\infty} \ln\left(1+\frac{1}{n^2}\right)$$

27. Determine the nature of the series using the p-Series Test:

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

28. Determine the nature of the series using the p-Series Test:

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

29. Determine the nature of the series using the p-Series Test:

$$\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$$

30. Determine the nature of the series using the p-Series Test:

$$\sum_{n=1}^{\infty} \frac{1}{n^{-3/2}}$$

Part II: Answers & Detailed Solutions

1. Series: $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{4^n}$

Solution: Rewrite the series as $\sum_{n=0}^{\infty} \left(-\frac{1}{4}\right)^n$. This is a geometric series with first term $a = 1$ and common ratio $r = -\frac{1}{4}$. Since $|r| = \frac{1}{4} < 1$, the series **converges**. Its sum is:

$$S = \frac{a}{1-r} = \frac{1}{1-\left(-\frac{1}{4}\right)} = \frac{1}{\frac{5}{4}} = \boxed{\frac{4}{5}}$$

2. Series: $\displaystyle\sum_{n=1}^{\infty} \left(1 - \frac{7}{4^n}\right)$

Solution: Applying the n-th term test for divergence:

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(1 - \frac{7}{4^n}\right) = 1 - 0 = 1 \neq 0$$

Since the limit of the n-th term is non-zero, the series **diverges** by the Test for Divergence.

3. Series: $\displaystyle\sum_{n=0}^{\infty} (-1)^n \frac{5}{4^n}$

Solution: Rewrite as $5\sum_{n=0}^{\infty} \left(-\frac{1}{4}\right)^n$. This is a geometric series with $a = 5$ and $r = -\frac{1}{4}$. Since $|r| = \frac{1}{4} < 1$, the series **converges**. Its sum is:

$$S = \frac{5}{1-\left(-\frac{1}{4}\right)} = \frac{5}{\frac{5}{4}} = \boxed{4}$$

4. Series: $\displaystyle\sum_{n=0}^{\infty} \left(\frac{5}{2^n} + \frac{1}{3^n}\right)$

Solution: Split into two convergent geometric series: $\sum_{n=0}^{\infty} \frac{5}{2^n} + \sum_{n=0}^{\infty} \frac{1}{3^n}$.

- First series: $a_1 = 5, r_1 = \frac{1}{2} \implies S_1 = \frac{5}{1-1/2} = 10$.
- Second series: $a_2 = 1, r_2 = \frac{1}{3} \implies S_2 = \frac{1}{1-1/3} = \frac{3}{2}$.

Total sum $S = 10 + \frac{3}{2} = \boxed{\frac{23}{2}}$. Since both component series converge, the total series **converges**.

5. Series: $1 + (-3) + (-3)^2 + (-3)^3 + \ldots$

Solution: Geometric series $\sum_{n=0}^{\infty}(-3)^n$ with $a = 1, r = -3$. Since $|r| = 3 \geq 1$, the series **diverges**.

6. Series: $\left(-\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)^3 + \left(-\frac{2}{3}\right)^4 + \ldots$

Solution: Geometric series starting at $n = 2$ with first term $a = \left(-\frac{2}{3}\right)^2 = \frac{4}{9}$ and common ratio $r = -\frac{2}{3}$. Since $|r| = \frac{2}{3} < 1$, it **converges**. Its sum is:

$$S = \frac{4/9}{1-\left(-\frac{2}{3}\right)} = \frac{4/9}{5/3} = \boxed{\frac{4}{15}}$$

7. Series: $\displaystyle\sum_{n=1}^{\infty} \frac{n}{n+10}$

Solution: Applying the n-th term test for divergence:

$$\lim_{n\to\infty} \frac{n}{n+10} = \lim_{n\to\infty} \frac{1}{1+10/n} = 1 \neq 0$$

Therefore, the series **diverges**.

8. **Series:** $\displaystyle\sum_{n=1}^{\infty} \frac{n}{n^2+3}$

Solution: Use the Limit Comparison Test with the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ (which diverges):

$$\lim_{n\to\infty} \frac{\frac{n}{n^2+3}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n^2}{n^2+3} = 1 > 0$$

Since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, the given series **diverges**.

9. **Series:** $\displaystyle\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$

Solution: Applying the n-th term test for divergence:

$$\lim_{n\to\infty} \cos\left(\frac{1}{n}\right) = \cos(0) = 1 \neq 0$$

Hence, the series **diverges**.

10. **Series:** $\displaystyle\sum_{n=0}^{\infty} \cos(n\pi)$

Solution: Note $\cos(n\pi) = (-1)^n$. The limit $\lim_{n\to\infty}(-1)^n$ does not exist (oscillates between 1 and -1, so $\neq 0$). By the Test for Divergence, the series **diverges**.

11. **Series:** $\displaystyle\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right)$

Solution: This is a telescoping series. The n-th partial sum is:

$$S_n = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1}$$

Taking the limit as $n \to \infty$: $\lim_{n\to\infty} S_n = 1 - 0 = 1$. The series **converges** to **1**.

12. **Series:** $\displaystyle\sum_{n=1}^{\infty} \left[\cos^{-1}\left(\frac{1}{n+1}\right) - \cos^{-1}\left(\frac{1}{n+2}\right)\right]$

Solution: Telescoping sum $S_n = \cos^{-1}\left(\frac{1}{2}\right) - \cos^{-1}\left(\frac{1}{n+2}\right)$. Taking the limit as $n \to \infty$:

$$\lim_{n\to\infty} S_n = \cos^{-1}\left(\frac{1}{2}\right) - \cos^{-1}(0) = \frac{\pi}{3} - \frac{\pi}{2} = -\frac{\pi}{\mathbf{6}}$$

The series **converges** to $-\pi/\mathbf{6}$.

13. **Series:** $\displaystyle\sum_{n=1}^{\infty} \left(\frac{3}{n^2} - \frac{3}{(n+1)^2}\right)$

Solution: Telescoping sum $S_n = 3\left(1 - \frac{1}{(n+1)^2}\right)$. Limit as $n \to \infty$:

$$\lim_{n\to\infty} S_n = 3(1-0) = 3$$

The series **converges** to **3**.

14. Series: $\displaystyle\sum_{n=1}^{\infty} \frac{4}{(4n-3)(4n+1)}$

Solution: Decompose using partial fractions: $\frac{4}{(4n-3)(4n+1)} = \frac{1}{4n-3} - \frac{1}{4n+1}$. The partial sum is $S_n = 1 - \frac{1}{4n+1}$. As $n \to \infty$, $S_n \to 1$. The sum is **1**.

15. Series: $\displaystyle\sum_{n=1}^{\infty} \frac{40n}{(2n-1)^2(2n+1)^2}$

Solution: Partial fraction expansion gives: $\frac{40n}{(2n-1)^2(2n+1)^2} = 5\left[\frac{1}{(2n-1)^2} - \frac{1}{(2n+1)^2}\right]$. Telescoping sum $S_n = 5\left(1 - \frac{1}{(2n+1)^2}\right)$. Taking limit as $n \to \infty$: $\lim_{n\to\infty} S_n = 5(1-0) = 5$. Sum is **5**.

16. Series: $\displaystyle\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right)$

Solution: Telescoping sum $S_n = 1 - \frac{1}{\sqrt{n+1}}$. Taking limit: $\lim_{n\to\infty} S_n = 1 - 0 = 1$. Sum is **1**.

17. Question: Can you make an infinite series of non-zero terms that converges to any number you want? Explain.

Solution: Yes. For any target real number $S \neq 0$, choose ratio r with $0 < |r| < 1$ (e.g. $r = 1/2$). Set first term $a = S(1-r)$ (e.g. $a = S/2$). Then $\sum_{n=0}^{\infty} ar^n$ has non-zero terms and converges to $\frac{a}{1-r} = \frac{S(1-r)}{1-r} = S$.

18. Question: If $\sum a_n$ converges and $a_n > 0$ for all n, can anything be said about $\sum \frac{1}{a_n}$?

Solution: Yes, $\sum \frac{1}{a_n}$ MUST DIVERGE. Since $\sum a_n$ converges, we must have $\lim_{n\to\infty} a_n = 0$. Since $a_n > 0$, taking the reciprocal yields $\lim_{n\to\infty} \frac{1}{a_n} = +\infty \neq 0$. By the Test for Divergence, $\sum \frac{1}{a_n}$ must **diverge to infinity**.

19. Question: Are there any values of $x \neq 0$ for which $\sum_{n=1}^{\infty} \frac{1}{nx}$ converges?

Solution: NO. There are NO non-zero values of x for which the series converges. Factor out $\frac{1}{x}$: $\sum_{n=1}^{\infty} \frac{1}{nx} = \frac{1}{x}\sum_{n=1}^{\infty} \frac{1}{n}$. The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges. Multiplying by non-zero constant $\frac{1}{x}$ yields a divergent series. Thus, it **diverges for all** $x \neq 0$.

20. Question: Is it true that if $\sum a_n$ is a divergent series of positive numbers, then there is also a divergent series $\sum b_n$ of positive numbers with $b_n < a_n$ for every n?

Solution: YES, IT IS TRUE. Define $b_n = \frac{1}{2}a_n$. Since $a_n > 0$, $b_n < a_n$ for every n. Furthermore, $\sum_{n=1}^{\infty} b_n = \frac{1}{2}\sum_{n=1}^{\infty} a_n$, which diverges. Thus, $\sum_{n=1}^{\infty} b_n$ is a strictly smaller divergent series.

21. Series: $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+30}$

Solution: Direct Comparison Test: $\frac{1}{n^2+30} < \frac{1}{n^2}$. The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a p-series with $p = 2 > 1$ (convergent). By Direct Comparison Test, the series **converges**.

22. Series: $\displaystyle\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^{3/2}}$

Solution: Since $0 \leq \cos^2 n \leq 1$, $\frac{\cos^2 n}{n^{3/2}} \leq \frac{1}{n^{3/2}}$. The p-series with $p = 3/2 > 1$ converges. By Direct Comparison Test, the series **converges**.

23. Series: $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n \cdot 3^n}$

Solution: For $n \geq 1$, $\frac{1}{n \cdot 3^n} \leq \frac{1}{3^n}$. Geometric series $\sum_{n=1}^{\infty} \frac{1}{3^n}$ converges ($r = 1/3 < 1$). By Direct Comparison Test, the series **converges**.

24. Series: $\displaystyle\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$

Solution: Limit Comparison Test with $b_n = \frac{1}{n}$: $\lim_{n\to\infty} \frac{\sin(1/n)}{1/n} = 1 > 0$. Harmonic series diverges, so $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$ also **diverges**.

25. Series: $\displaystyle\sum_{n=2}^{\infty} \frac{n(n+1)}{(n^2+1)(n-1)}$

Solution: For large n, $a_n = \frac{n^2+n}{n^3-n^2+n-1} \approx \frac{1}{n}$. Limit Comparison Test with $b_n = 1/n$ yields limit $1 > 0$. Since $\sum_{n=2}^{\infty} \frac{1}{n}$ diverges, the given series **diverges**.

26. Series: $\displaystyle\sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n^2}\right)$

Solution: Using inequality $\ln(1+x) \leq x$ for $x > 0$: $\ln\left(1 + \frac{1}{n^2}\right) \leq \frac{1}{n^2}$. p-series with $p = 2 > 1$ converges. By Direct Comparison Test, the series **converges**.

27. Series: $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

Solution: Rewrite as $\sum \frac{1}{n^{1/2}}$. This is a p-series with $p = 1/2 \leq 1$. Therefore, the series **diverges**.

28. Series: $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$

Solution: This is a p-series with $p = 2 > 1$. Therefore, the series **converges**.

29. Series: $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$

Solution: This is a p-series with $p = 3/2 > 1$. Therefore, the series **converges**.

30. Series: $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{-3/2}}$

Solution: Rewrite as $\sum n^{3/2}$. Limit as $n \to \infty$ is $\infty \neq 0$. By Test for Divergence (or p-series with $p = -3/2 \leq 1$), the series **diverges**.